Exercise 1 (Polynomial Regression):

When examining the scatter plot (Fig. 1) of an individual's wage against their age, the effect of the age seems to be rather quadratic than linear. We now want to carry out a polynomial regression assuming a second-order polynomial for the effect of age.

(a) Specify the model equation.

(b) According to the method of least squares, the parameter estimates for the model are $\hat{\beta}_0 = -10.43, \hat{\beta}_1 = 5.29$ and $\hat{\beta}_2 = -0.05$. The estimated regression line is displayed in Fig. 2. Interpret the results with the help of Fig. 2.

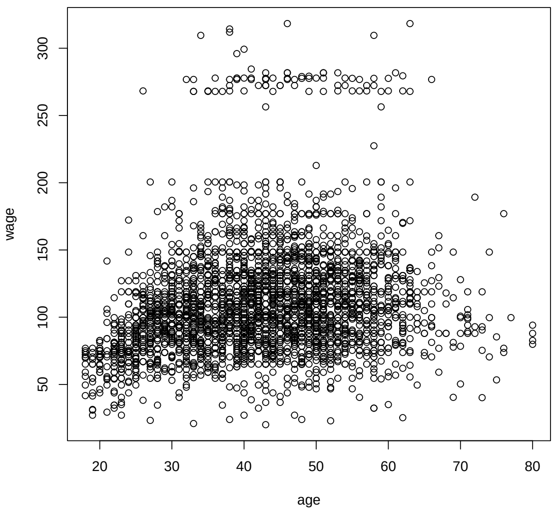

Figure 1: Scatter plot of wage and age

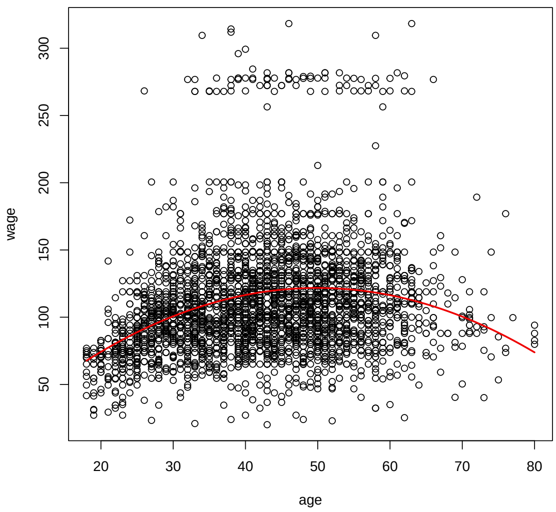

Figure 2: Scatter plot with estimated effect

Exercise 2 (Interaction between Covariates):

We now want to model the effect of an individual's age and health on his wage as well as taking an interaction effect between age and health into account.

(a) Define the dummy variable for health and specify the model equation.

(b) The method of least squares gave the following fitted model:

$$\widehat{wage} = 78.66 + 0.51 \cdot age - 1.81 \cdot health.Verygood + 0.43 \cdot age \cdot health.Verygood$$

Calculate the estimated intercept and slope for a man with less than good health and for a man with health greater than very good.

(c) Figure 3 shows the estimated regression lines for men with health level less than good and for men with health level greater than very good for the model without an interaction term between age and health. Here, the fitted model is as follows:

$$\widehat{wage} = 65.74 + 0.80 \cdot age + 16.90 \cdot health.Verygood.$$

Figure 4 shows the estimated regression lines for the model from (a) and (b). What does the interaction term do from a geometric perspective?

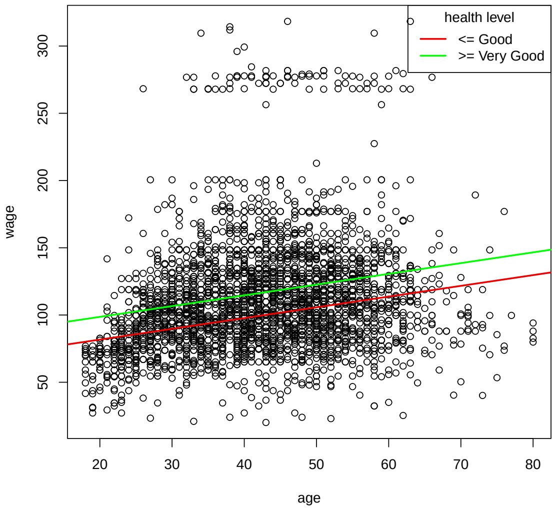

Figure 3: Estimated regression lines for the model without an interaction term

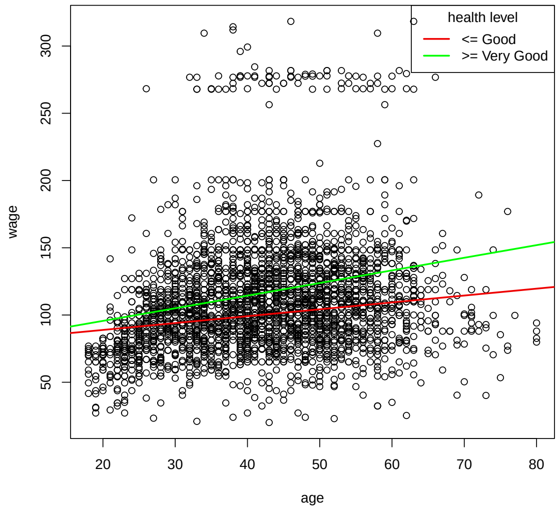

Figure 4: Estimated regression lines for the model with an interaction term